THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 11: **Exploring Some Geometric Themes** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. All the 12 faces of a polyhedron are pentagons. Its number of vertices is: 1

(a) 20 **(b)** 30 **(c)** 44 **(d)** 18

2. A polyhedron has 18 edges and exactly 3 edges meet at every vertex. Its number of faces is: 1

(a) 20 **(b)** 6 **(c)** 12 **(d)** 8

3. If every corner of a cube is sliced off (without removing the whole edges), the number of vertices of the resulting solid is: 1

(a) 24 **(b)** 12 **(c)** 8 **(d)** 16

4. A cube is painted all over and cut into unit cubes. If exactly 24 of the unit cubes have exactly one face painted, the side of the big cube is: 1

(a) 4 **(b)** 3 **(c)** 5 **(d)** 6

5. A Koch curve starts with a 81 cm line. The length of the curve at stage 4 (in cm) is: 1

(a) 337 **(b)** 324 **(c)** 256 **(d)** 229

6. A Sierpinski triangle starts with a triangle of side 64 cm. At stage 4 the side of each small shaded triangle is (in cm): 1

(a) 8 **(b)** 2 **(c)** 16 **(d)** 4

7. A polyhedron has 8 triangular faces and 2 quadrilateral faces and no other faces. Its number of vertices is: 1

(a) 10 **(b)** 16 **(c)** 28 **(d)** 8

8. All the 16 faces of a polyhedron are pentagons. Its number of vertices is: 1

(a) 58 **(b)** 40 **(c)** 24 **(d)** 26

9. **Assertion (A):** A cube satisfies Euler's formula. 1

Reason (R): For every convex polyhedron, $F + V - E = 2$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** A triangular prism has 5 faces. 1

Reason (R): A pyramid with an n-sided base has $n + 1$ faces.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A cube is painted all over and cut into unit cubes. If 48 unit cubes have exactly two painted faces, find the side of the cube and the total number of unit cubes. 2

12. A polyhedron has 8 triangular faces and 6 square faces only. Find its number of edges. 2

13. A Sierpinski triangle starts with a triangle of side 128 cm. Find the number of shaded triangles and the side of each at stage 5. 2

14. A cube is painted all over and cut into unit cubes. If 84 unit cubes have exactly two painted faces, find the side of the cube and the total number of unit cubes. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Every face of a polyhedron is a triangle and it has 32 faces. Find the number of edges and vertices and verify Euler's formula. 3

16. A cube is painted all over and cut into unit cubes. For which side is the number of unit cubes with exactly two painted faces equal to 144? Hence find the number with exactly one painted face. 3

17. In the Sierpinski triangle construction, find the first stage at which the number of shaded triangles exceeds 500. How many shaded triangles are there at that stage? 3

OR

Every face of a polyhedron is a triangle and it has 40 faces. Find the number of edges and vertices and verify Euler's formula. 3

18. A cube is painted all over and cut into unit cubes. For which side is the number of unit cubes with exactly two painted faces equal to 168? Hence find the number with exactly one painted face. 3

OR

In the Sierpinski triangle construction, find the first stage at which the number of shaded triangles exceeds 1000. How many shaded triangles are there at that stage? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Rohan makes solid models of a pentagonal prism and a pentagonal pyramid for a school exhibition and counts their faces, vertices and edges. 5

(i) How many faces does the pentagonal prism have?
(ii) How many vertices does it have?
(iii) How many edges does it have?
(iv) Verify Euler's formula for the prism and for the pentagonal pyramid.

20. A polyhedron has 12 triangular faces and 4 quadrilateral faces and no other faces. Find the number of edges and vertices, and verify Euler's formula. 5

OR

A Koch snowflake is built from an equilateral triangle of side 9 cm. Find its perimeter at stage 2, and by how much the perimeter has increased from stage 0. 5

END OF QUESTION PAPER — Total: 40 Marks